

Advanced (Habanero)

Q1. What is the vertex of the graph of $y = |x - 2|$?

- (A) (2, 0)
- (B) (0, 2)
- (C) (2, 2)
- (D) (0, 0)
- (E) (1, 1)

Q2. The graph of $y = |x + 1|$ has which axis of symmetry?

- (A) $x = -1$
- (B) $x = 0$
- (C) $x = 1$
- (D) $y = -1$
- (E) $y = 0$

Q3. Which of the following is the graph of $y = 2|x|$?

- (A) A graph with vertex at (0, 0) and opening upwards
- (B) A graph with vertex at (0, 0) and opening downwards
- (C) A graph with vertex at (0, 2) and opening upwards
- (D) A graph with vertex at (0, -2) and opening downwards
- (E) A graph with vertex at (2, 0) and opening to the right

Q4. What is the direction of opening for the graph of $y = -|x - 1|$?

- (A) Upwards
- (B) Downwards
- (C) To the left
- (D) To the right
- (E) None of the above

Q5. The graph of $y = |x| - 2$ has which vertex?

- (A) (0, -2)
- (B) (0, 0)
- (C) (0, 2)
- (D) (2, 0)
- (E) (-2, 0)

Q6. Which of the following graphs represents the equation $y = |x + 2| + 1$?

- (A) A graph with vertex at (-2, 1) and opening upwards
- (B) A graph with vertex at (-2, 1) and opening downwards
- (C) A graph with vertex at (2, 1) and opening upwards
- (D) A graph with vertex at (2, 1) and opening downwards
- (E) A graph with vertex at (0, 1) and opening to the right

Q7. The equation $y = 2|x - 3|$ has which axis of symmetry?

- (A) $x = 3$
- (B) $x = 0$
- (C) $x = -3$
- (D) $y = 2$
- (E) $y = -2$

Q8. What is the vertex of the graph of $y = -2|x + 1| + 3$?

- (A) (-1, 3)
- (B) (0, 3)
- (C) (1, 3)
- (D) (-1, 0)
- (E) (0, 0)

Open Ended Questions (FRQ)**Q9.** Graph the function $y = |x - 2| + 1$. Identify the vertex, axis of symmetry, and direction of opening.**Q10.** Compare and contrast the graphs of $y = |x|$ and $y = |x + 2| + 1$. How do the vertices, axes of symmetry, and directions of opening differ?**Chef's Tips**

- Tip 1: Ensure students understand the properties of absolute value functions and their graphs.
- Tip 2: Provide examples of transformations and how they affect the graph.
- Tip 3: Encourage students to use graphing tools or software to visualize the functions.